

THE UNIVERSITY OF
MELBOURNE

FNCE 30007
Derivative Securities

Lecture 6 – Binomial Model II

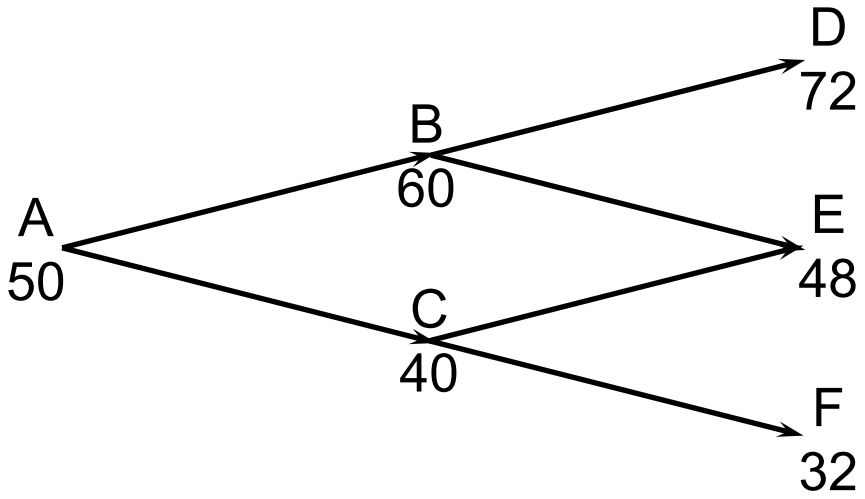
Dr Xiaoting Wei

Outline

- Recap of Week 5 Lecture
- Binomial Model Applied to American Options
- Multi-period Binomial Model

Readings: Chapter 12

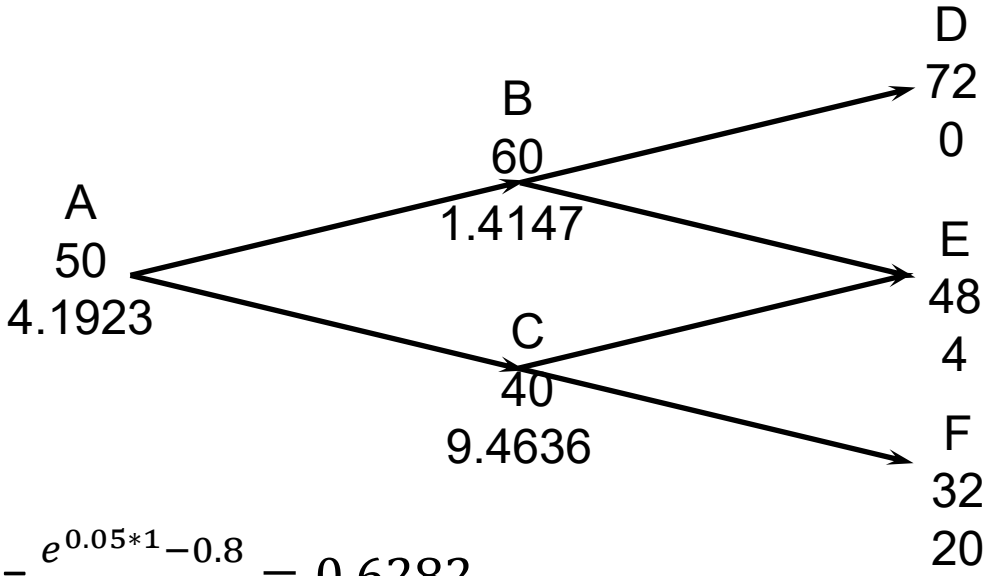
Recap of Week 5 Lecture (slide 33)



Valuing a put option using a two-step binomial model

- $K = 52$
- time step =1yr
- $r = 5\%$
- $u = 1.2$, $d = 0.8$

Recap of Week 5 Lecture (slide 33)



$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.05*1} - 0.8}{1.2 - 0.8} = 0.6282$$

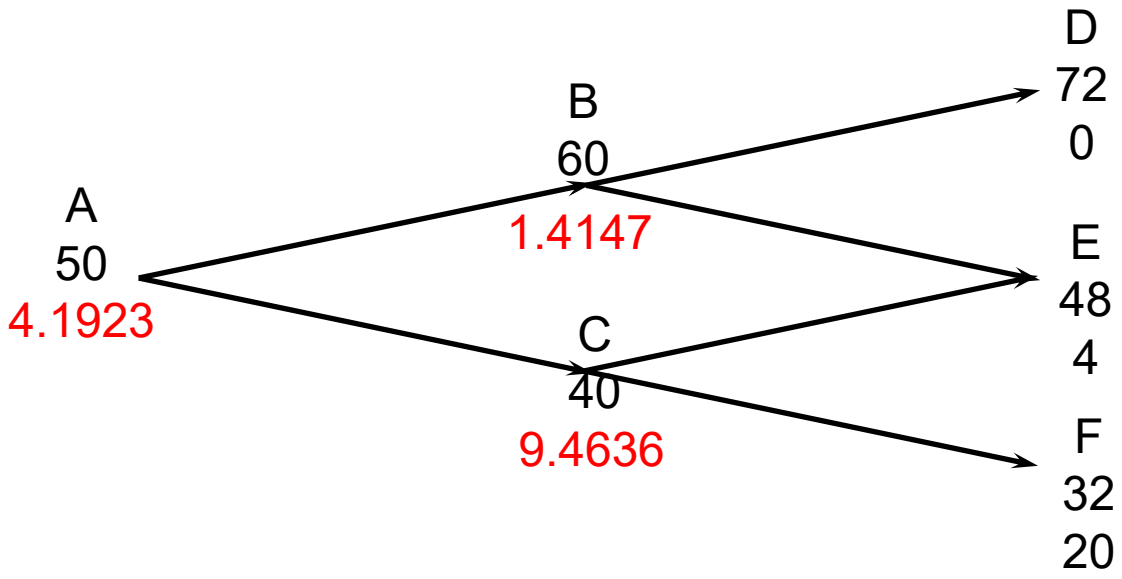
- Node B: $f_u = (p * f_{uu} + (1 - p) * f_{ud}) * e^{-rT}$
- $f_u = (0.6282 * 0 + (1 - 0.6282) * 4) * e^{-0.05*1} = 1.4147$
- Node C: $f_d = (0.6282 * 4 + (1 - 0.6282) * 20) * e^{-0.05*1} = 9.4636$
- Node A: $f = (0.6282 * 1.4147 + (1 - 0.6282) * 9.4636) * e^{-0.05*1}$

$= 4.1923$

Alternatively, $f = \frac{p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}} = 4.1923$

Binomial Model Applied to American Options

- In the Week 5 lecture, all the options we have considered have been European options. We now move on to **AMERICAN OPTIONS**

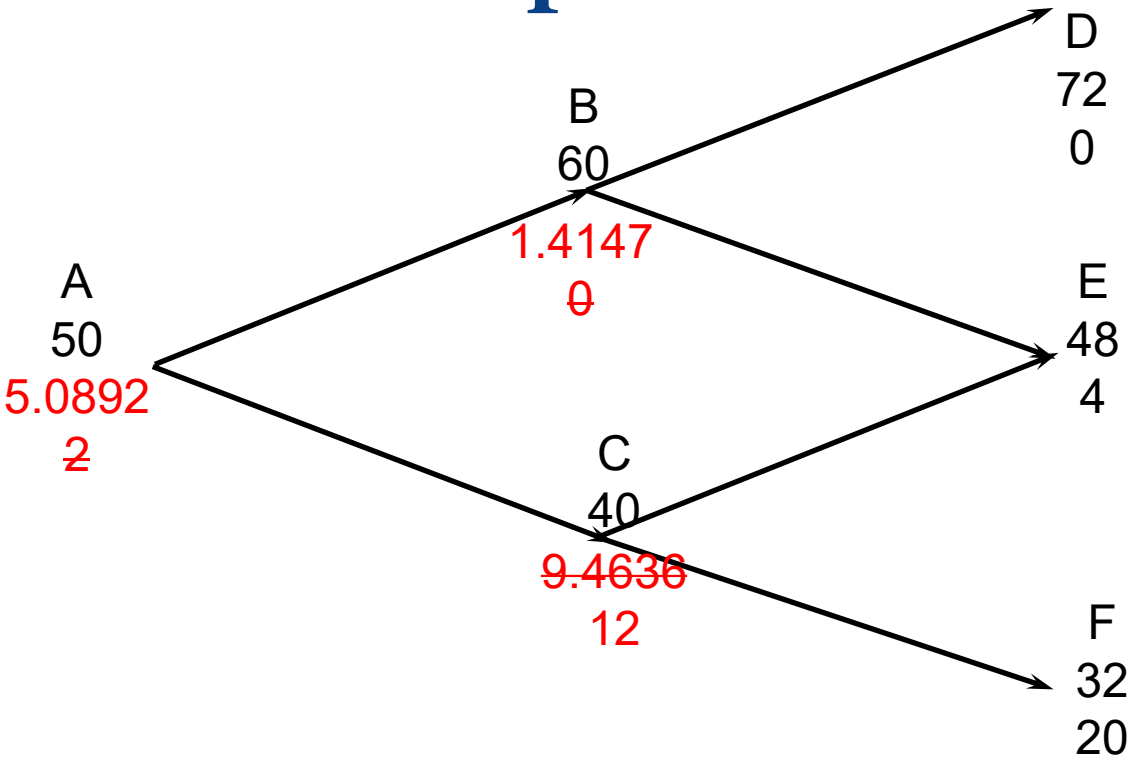


- With American options, early exercise is possible!

Binomial Model Applied to American Options

- No analytical formula!
- Start at maturity & work backwards through the stock price tree
- The value at the final nodes is the same as for the European option
- At earlier nodes the value of the option is the greater of:
  - The value of a European option at that node
  - The payoff from early exercise

Binomial Model Applied to American Options



B: Early exercise, intrinsic value = 0 since 52-60=-8
No early exercise, value = 1.4147 (time value) ✓
C: Early exercise, intrinsic value = 52-40 =12 ✓
No early exercise, value = 9.4636
A: Early exercise, intrinsic value = 52-50=2
No early exercise, value = [0.6282x1.4146+(1-0.6282)x12]/1.0513 = 5.0892 ✓

Binomial Model Applied to American Options

- When early exercise may be optimal

  i) American call on non div paying stock should never be exercised early

  - If exercise, receive intrinsic value
  - If wait, receive intrinsic value + time value

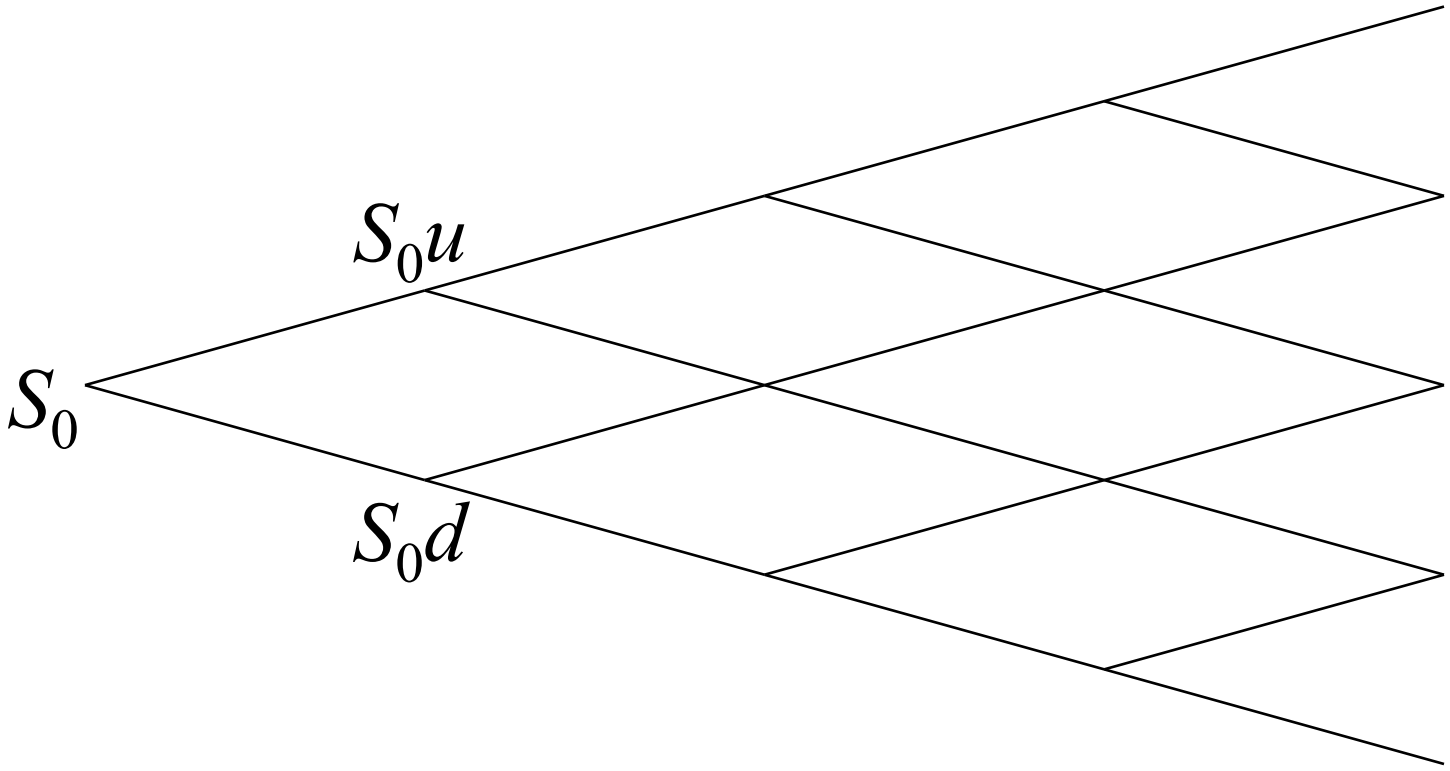
  ii) American put on a non div paying stock

  - May be optimal to exercise early (if so time value =0)
  - Discussion in lecture 4

Multi-period Binomial Model

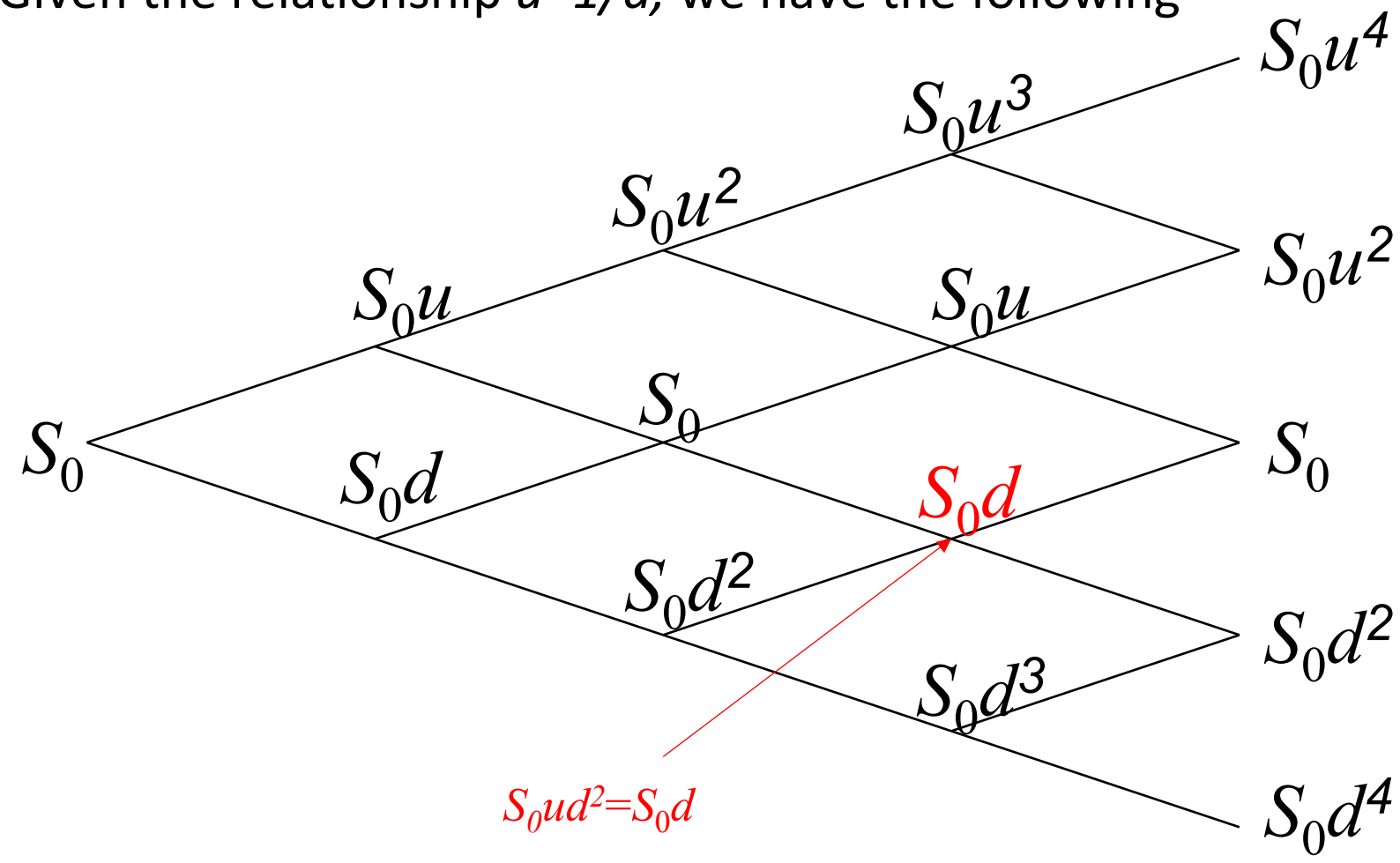
Let *i* be the number of steps in a path, and *j* be the number of up movements. Therefore, there are *i-j* down movements in a path. At time $i\Delta t$, there are *i+1* stock prices.

The price at a node is $S_0 u^j d^{i-j}$, where *j=0,1,…i*



Multi-period Binomial Model

Given the relationship *u=1/d,* we have the following



Multi-period Binomial Model

An American put option with

S_0 = 50; K = 50; r =10%; σ = 40%;

T = 5 months = 0.4167;

Δt = 1 month = 0.0833

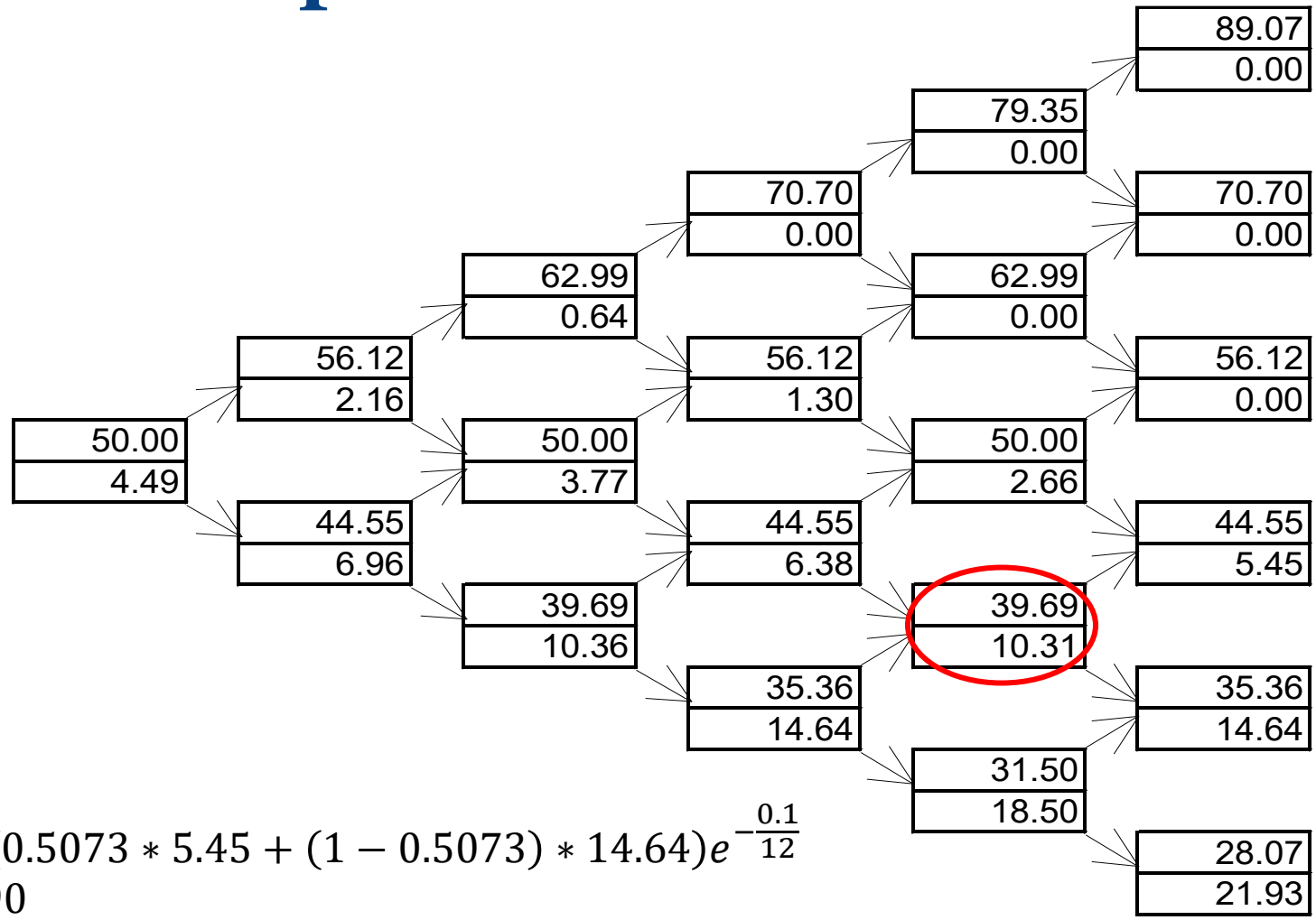
The parameters imply

$$u = e^{\sigma\sqrt{\Delta t}} = 1.1224$$

$$d = \frac{1}{u} = 0.8909$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = 0.5073$$

Multi-period Binomial Model



$f = (0.5073 * 5.45 + (1 - 0.5073) * 14.64)e^{-\frac{0.1}{12}}$
$= 9.90$

Intrinsic value = 50-39.69 = 10.31 ✓

Introducing the simulator

Let’s go to the LMS and find the simulator